

assignment #01

Course: CHE 301: CHEMICAL ENGINEERING THERMODYNAMICS
Term: Spring 2026
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Due Date: Friday, January 16th TA's Mailbox by 3:00PM

INSTRUCTIONS FOR ASSIGNMENT DELIVERY

- Prepare a typed report of your assignment.
- Include this cover page with your name(s) in the report.
- Upload your report as a PDF file by the Due Date and Time on the UIC Blackboard.
- Place a hard copy of your report as a PDF file by the Due Date and Time in the TA's mailbox.

- Assignments that are not typed will not be accepted.
- Late Assignments will not be accepted without prior approval.
- You may use the Internet as well as ChatGPT or other public tools as long as you place a written statement about their use in your assignment report

Abstract

This report presents the analysis of two independent thermodynamic and numerical problems. Problem A examines the performance of an ideal Brayton cycle operating under steady-state conditions with constant heat capacity and ideal gas behavior. The compressor pressure ratio that maximizes net power output is derived analytically and used to evaluate cycle temperatures, thermal efficiency, Carnot efficiency, network output, and the minimum air flow rate required to produce 50 MW of power. Problem B focuses on the numerical solution of the Rachford–Rice equation for vapor–liquid equilibrium. The vapor fraction of a multicomponent hydrocarbon mixture is computed using the Newton method and the Secant method, with results verified using Excel’s Goal Seek tool. The results demonstrate consistency between analytical derivations, numerical methods, and thermodynamic limits

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Introduction and Background

Thermodynamic cycles and numerical solution techniques are fundamental tools in chemical engineering analysis. The Brayton cycle represents an idealized gas turbine power cycle and is commonly used to evaluate the limits of power generation efficiency. Separately, phase equilibrium calculations are essential for the design and operation of separation processes, where numerical methods are often required to solve nonlinear algebraic equations such as the Rachford–Rice equation. This report integrates both classical thermodynamics and numerical methods to reinforce these core concepts.

Problem A: Ideal Brayton Cycle Analysis

Cycle Description and Assumptions

$$P_1 = 100 \text{ kPa}$$

$$T_1 = 300 \text{ K}$$

$$T_3 = 1400 \text{ K}$$

$$P_4 = P_1$$

Assumptions:

Steady-state operation

Ideal gas behavior

Constant molar heat capacity

Isentropic compression and expansion

Negligible kinetic and potential energy changes

$$\gamma = 1.4$$

$$C_p = 29.1 \text{ J mol}^{-1} \text{ K}^{-1}$$

Compressor pressure ratio

$$r_p = P_2 / P_1$$

1) Maximum net work

$$T_2 / T_1 = T_3 / T_4$$

$$T_2 / T_1 = r_p^{(\gamma-1)/\gamma}$$

$$T_4 / T_3 = r_p^{-(\gamma-1)/\gamma}$$

$$r_p^{(\gamma-1)/\gamma} = \sqrt{T_3 / T_1}$$

$$r_{p,opt} = (T_3 / T_1)^{\gamma / (2(\gamma-1))}$$

$$r_{p,opt} = (1400 / 300)^{(1.4 / (2(0.4)))} = (4.667)^{1.75}$$

$$r_{p,opt} = 14.8$$

2) Compressor discharge pressure

$$P_2 = r_p P_1 = 14.8 \times 100 = 1480 \text{ kPa}$$

Compressor discharge temperature

$$T_2 = T_1 r_p^{(\gamma-1)/\gamma}$$

$$T_2 = 300 \times (14.8)^{0.286} = 612 \text{ K}$$

Turbine exhaust temperature

$$T_4 = T_3 / r_p^{(\gamma-1)/\gamma}$$

$$T_4 = 1400 / (14.8)^{0.286} = 686 \text{ K}$$

3) Thermal efficiency at maximum net power

Heat added:

$$q_{in} = C_p (T_3 - T_2)$$

Heat rejected:

$$q_{out} = C_p (T_4 - T_1)$$

Thermal efficiency:

$$\eta_{th} = 1 - (T_4 - T_1) / (T_3 - T_2)$$

$$\eta_{th} = 1 - (686 - 300) / (1400 - 612)$$

$$\eta_{th} = 0.51 = 51\%$$

4) Carnot efficiency

$$\eta_{Carnot} = 1 - T_{low} / T_{high} = 1 - 300 / 1400$$

$$\eta_{Carnot} = 0.786 = 78.6\%$$

Comparison

Brayton (ideal) = 51%

Carnot (upper limit) = 78.6%

Brayton is lower as expected

5) Net work at maximum power (per mole)

$$w_{net} = C_p [(T_3 - T_4) - (T_2 - T_1)]$$

$$w_{net} = 29.1 [(1400 - 686) - (612 - 300)]$$

$$w_{net} = 29.1 (714 - 312) = 29.1 \times 402$$

$$w_{net} = 11.7 \text{ kJ/mol}$$

6) Air flow rate for 50 MW (in MMSCFD)

$$\dot{n} = (50 \times 10^6) / (11.7 \times 10^3) = 4270 \text{ mol/s}$$

$$1 \text{ lbmol} = 379 \text{ scf}$$

$$1 \text{ lbmol} = 453.6 \text{ mol}$$

$$\dot{n} = 4270 / 453.6 = 9.41 \text{ lbmol/s}$$

$$\dot{V} = 9.41 \times 379 \times 86400$$

$$\dot{V} = 308 \text{ MMSCFD}$$

B) Numerical Solution of the Rachford-Rice Equation

Governing Equation

The vapor fraction β of a two-phase hydrocarbon mixture is defined by:

$$f(\beta) = \sum [z_i (K_i - 1) / (1 + \beta (K_i - 1))] = 0$$

where:

z_i = molar composition

K_i = equilibrium ratio

β = vapor fraction

1. Analytical Derivative

The derivative of $f(\beta)$ with respect to β is:

$$df/d\beta = - \sum [z_i (K_i - 1)^2 / (1 + \beta (K_i - 1))^2]$$

2. Newton Method

Newton iteration formula:

$$\beta_{k+1} = \beta_k - f(\beta_k) / f'(\beta_k)$$

Initial guess:

$$\beta_0 = 0.5$$

Newton Method Iteration Table

Iteration	β	$f(\beta)$
1	0.500000	-1.223288
2	0.077671	0.370114
3	0.105756	0.076169
4	0.114865	0.004488
5	0.115470	1.71E-05
6	0.115473	2.52E-10

Newton method converges to:

$$\beta = 0.11547$$

3. Secant Method

Secant iteration formula:

$$\beta_{k+1} = \beta_k - f(\beta_k) (\beta_k - \beta_{k-1}) / [f(\beta_k) - f(\beta_{k-1})]$$

Initial guesses:

$$\beta_0 = 0.30$$

$$\beta_1 = 0.60$$

Secant Method Iteration Table

Iteration	β	$f(\beta)$
1	0.600000	-1.555514
2	0.046664	0.953852
3	0.256996	-0.598104
4	0.175937	-0.328621
5	0.077089	0.377836
6	0.129956	-0.097748

Further iterations converge to:

$$\beta \approx 0.1155$$

Excel GoalSeek converges to:

$$\beta = 0.11547$$

$$\beta = 0.11547$$

Conclusion

In Problem A, the optimal compressor pressure ratio for maximum net power output in an ideal Brayton cycle was analytically determined as 14.8. At this condition, the cycle achieves a thermal efficiency of 51%, which is lower than the 78.6% Carnot efficiency, as dictated by the second law of thermodynamics. The required air flow rate to produce 50 MW of power was calculated to be 308 MMSCFD.

In Problem B, the vapor fraction of a multicomponent hydrocarbon mixture was successfully computed using both the Newton and Secant methods. All numerical approaches, including Excel GoalSeek, converged to the same solution of $\beta = 0.11547$, confirming the correctness and stability of the numerical solution.